

TAOD-Lite: Task-Aware Object Detection, Lightweight

Stage 2A Report – DVCon India 2026 Design Contest

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Problem Statement

TAOD-Lite addresses task-aware object selection: given a scene image and one everyday task ID from the COCO-Tasks benchmark, the system detects objects and ranks which detected object is most suitable for performing the task. The Stage 2A objective is functional correctness under CPU-only inference, while keeping the pipeline lightweight and compatible with future hardware-oriented deployment.

Approach and Pipeline Architecture

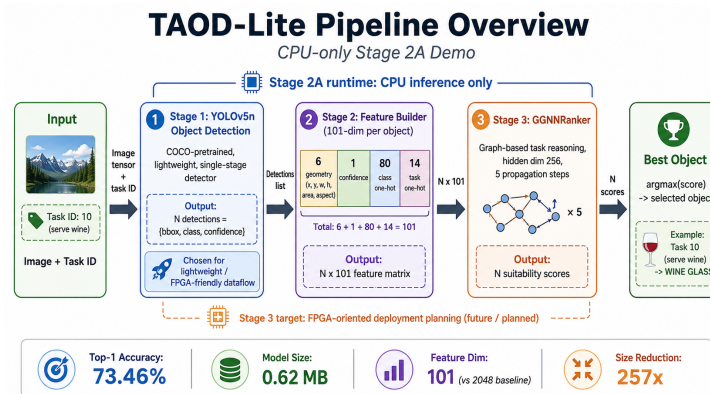


Figure 1: End-to-end TAOD-Lite pipeline: Image + Task ID → YOLOv5n detections → $N \times 101$ features → GGNNRanker scores → best object.

Stage 1 – YOLOv5n Object Detection. The detector is a pretrained YOLOv5n model trained on COCO categories. It replaces the Faster R-CNN detector used in the original COCO-Tasks pipeline with a compact, single-stage detector that produces bounding boxes, class IDs, and confidence scores. YOLOv5n was selected because its fixed-grid, deterministic dataflow is more suitable for later FPGA-oriented optimization than a heavier region-proposal detector.

Stage 2 – Node Feature Construction. Each detected object is converted into a 101-dimensional node feature vector: 6 geometry features ($c_x, c_y, w, h, \text{area}, \log\text{-aspect-ratio}$), 1 confidence value, an 80-dimensional COCO class one-hot vector, and a 14-dimensional task one-hot vector. This replaces the 2048-dimensional ResNet ROI feature representation, giving a 95.1% per-node feature reduction.

Stage 3 – GGNN Task Ranking. The GGNNRanker projects each 101-dimensional node to a 256-dimensional hidden representation, performs fully connected graph message passing for 5 propagation steps using a GRU update, and applies a score head to output one suitability score per detection. The selected answer is the detection with the highest score for the requested task.

Training Methodology

The ranker was trained on 18,133 COCO-Tasks samples with a split of 13,590 training, 1,510 validation, and 3,033 test samples. Training used 80 epochs, AdamW optimization with learning rate 3×10^{-4} and weight decay 1×10^{-5} , ReduceLROnPlateau scheduling, CrossEntropyLoss, batch size 32, gradient clipping at max norm 2.0, and random seed 42.

Stage 1 Modifications from the Original Proposal. Small Stage 2A adjustments were made for reliable demonstration: inference is forced to CPU with `DEVICE = 'cpu'` and `torch.set_num_threads(4)`, and the YOLO confidence threshold is lowered to 0.15 to improve recall on small or partially occluded objects. These changes preserve the task-ranking pipeline while improving demo coverage.

Stage 2A Compliance and Scope. All Stage 2A inference is executed on CPU only; GPU use is limited to training. The current submission does not claim FPGA implementation results. A known limitation is that YOLOv5n, as a nano detector, may miss small or occluded task-critical objects such as a shovel, carpet, or bottle; when this occurs, the GGNN correctly ranks only among the objects available from detection. This is a lightweight detection-coverage tradeoff, not a GGNN logic failure.

Results and Demonstrations

Top-1 Accuracy

73.46%

Model Size

0.62 MB

Speedup vs Baseline

13.5×

Size Reduction

257× smaller

Test loss: 0.769 — Test samples: 3,033 — Training samples: 13,590 — Validation samples: 1,510 — Input feature dim.: 101

Testing Snapshots

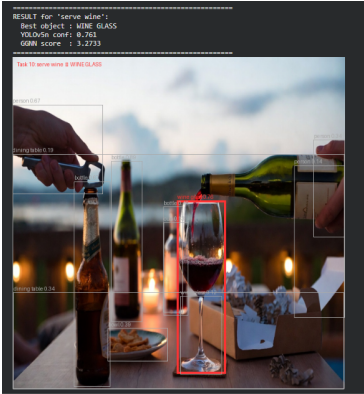


Figure 2: Task 10, serve wine: restaurant scene. The system selects **WINE GLASS** with GGNN score +3.35 and confidence 0.818.

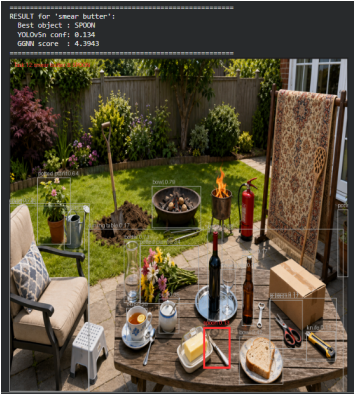


Figure 3: Secondary CPU-only test snapshot. Use the Task 12 smear-butter result or the Task 5 water-a-plant result here.

Result Summary

Metric	Value
Test Top-1 Accuracy	73.46%
Test Loss	0.769
Model Size	0.62 MB compared with 160 MB Faster R-CNN + GGNN baseline
Size Reduction	257× smaller
Pipeline Latency	9.4 ms / 106.1 FPS from original T4 GPU measurement; Stage 2A inference is run on CPU only
Speedup over Baseline	13.5× compared with 127.33 ms / 7.85 FPS baseline
Feature Dimension	101, reduced from the 2048-dimensional baseline representation

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Image: Image1.png
Detected: ['bowl', 'bottle', 'chair', 'potted plant', 'couch', 'bowl', 'cup', 'potted plant', 'vase']
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Task	Task Name	Best Object	Score
1	step on something	chair	3.2556
2	sit comfortably	couch	3.3831
3	place flowers in	bottle	-0.0543
4	get potatoes out of fire	bowl	0.8075
5	water a plant	bowl	1.0914
6	get lemon out of tea	bottle	0.2266
7	dig a hole	bowl	0.7193
8	open a bottle of beer	chair	2.9273
9	open a parcel	bottle	-5.3599
10	serve wine	cup	0.0834
11	pour sugar	cup	0.7637
12	smear butter	cup	-1.8559
13	extinguish fire	bottle	0.5334
14	pound a carpet	bowl	0.9200

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results

Interpretation. TAOD-Lite achieves 73.46% test Top-1 accuracy with a compact 0.62 MB ranker, compared with the 160 MB Faster R-CNN + GGNN baseline, giving a 257× model-size reduction. The wine-glass example is the clearest validation case: when the correct task-relevant object is detected, the GGNN ranks it correctly for the task. Remaining failures are mainly caused by YOLOv5n recall limits, which is acceptable for Stage 2A and directly tied to the lightweight CPU-only and future hardware-oriented design choice.